JOB MARKET SKILLS AND SALARY ANALYSIS

Data Analytics Project Report

Project ID	DA-021
Group ID	BT_DA_03
Program	B.Tech Intro to Data Analytics
Section	A
Mentor	Arsh Singh
Team Size	14

A structured analysis of job-market data to understand salary patterns, skill demand, experience levels, job roles, locations, and the relationship between skills and salary.

1. Introduction

The Job Market Skills and Salary Analysis project focuses on cleaning and analyzing job-market data to understand skill demand and salary patterns. The analysis brings together job role, experience, location, technical skills, company information, and salary data to create a clearer view of the market.

The project is implemented as a Python/Colab-based data analytics workflow. The notebook uses Pandas and NumPy for data handling and Matplotlib/Seaborn for visualization in the source workflow. The resulting analysis is intended to help students and job seekers understand market trends and prioritize relevant skills.

2. Objectives

- Analyze salary patterns across different job roles and career levels.
- Identify highly demanded technical skills from job listings.
- Compare salary levels across Entry, Mid, and Senior experience levels.
- Analyze location-wise differences in job availability and salary.
- Study the relationship between technical skills and salary.
- Analyze correlations between years of experience, skill count, and salary.
- Present the results through clear visualizations for easier comparison.
- Provide useful insights for students preparing for the job market.

3. Tools and Technologies

Tool / Technology	Purpose
Python	Core programming language for analysis
Pandas	Data loading, cleaning, grouping and analysis
NumPy	Numerical data processing
Matplotlib	Data visualization
Seaborn	Statistical and comparative visualization
Google Colab	Notebook-based execution environment
CSV Dataset	Source of job-market records

4. Dataset and Data Preparation

The notebook loads the file job_market_skills_salary_analysis.csv into a Pandas DataFrame. The dataset is inspected using shape, information, and descriptive-statistics operations before analysis.

Data Cleaning Steps

- Missing values in the categorical columns Education, Company, and Skills are filled using the most frequent value (mode).
- Duplicate rows are removed and the index is reset.
- The cleaned dataset is then used for salary, skill, role, location, and correlation analysis.

5. Analysis Performed

Basic Salary Analysis — Average, median, and highest annual salary are calculated. Salary is also grouped by Job Role.

Most Demanded Skills — The Skills field is split and counted to identify the most frequently occurring skills.

Salary by Experience — Average salary is compared across experience levels.

Highest-Paying Job Roles — Average salary is calculated for each job role and compared.

Location-wise Salary — Average salary is grouped by Location to compare market differences.

Skills and Salary — Each skill is associated with job count and average salary.

Correlation Analysis — Relationships among Years_of_Experience, Skill_Count, and Annual_Salary_INR are examined.

6. Key Findings and Interpretation

- The analysis allows comparison of the most demanded technical skills using skill-count tables and charts.
- Average salary differences can be examined across job roles and experience levels.
- Location-wise salary analysis helps identify locations with comparatively higher average salary levels.
- The skill-salary table helps connect technical skills with associated job counts and average salary.
- Correlation analysis provides a broader view of how experience, number of skills, and salary vary together.
- The combined analysis shows that job-market patterns are influenced by multiple factors rather than a single variable.

7. Practical Relevance

The project can be useful for students, fresh graduates, and job seekers who want to understand which skills appear frequently in job listings and how salary patterns vary with role, experience, and location. The analysis can support more informed skill-development and career-planning decisions.

8. Future Scope

- Add more recent job-market records to improve the breadth of the analysis.
- Build an interactive dashboard using Plotly, Power BI, or Streamlit.
- Add predictive models for salary estimation.
- Perform time-based analysis when historical job-posting data is available.
- Use NLP techniques to extract and standardize skills automatically from job descriptions.

9. Conclusion

The Job Market Skills and Salary Analysis provides a structured view of how job role, experience, location, and technical skills can be explored together to understand job-market demand and salary patterns. The workflow demonstrates the practical use of data cleaning, grouping, aggregation, skill counting, and correlation analysis in a real-world career-data context.

Overall, the project turns raw job-market records into meaningful analytical insights that can help students and job seekers better understand current market requirements and plan their technical skill development.

“Understand the market. Build the right skills. Prepare for the future.”

10. Team Members

S. No.	Student Name	Role
1	Prince Yadav	Team Member
2	Himanshu Singh	Team Member
3	Inderveer Kaur	Team Member
4	Himanshu Patel	Team Member
5	Syeda Aneeqa Adil	Team Member
6	Mohammad Atif	Team Member
7	Mohammad Ubaid	Team Member
8	Anuj Kumar	Team Member
9	Nitin	Team Member
10	Kapil Sagar	Team Member
11	Altamash	Team Member
12	Arun Kashyap	Team Member
13	Khanak Agarwal	Team Member
14	Alima Shan	Team Member

Note: All 14 students are listed as Team Members; no Team Leader designation is used.

11. Project Information

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